
Unsupervised Discovery of Gendered Rhetoric: A Non-Negative Matrix Factorization Analysis of French Electoral Manifestos (1973–1993)

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Abstract

In this project, I quantify the thematic divergence between male and female political candidates in France during the legislative elections from 1973 to 1993. Leveraging a corpus of 20,331 digitized electoral manifestos, I employ Non-negative Matrix Factorization (NMF) to identify latent rhetorical structures that characterize gendered political discourse. While the exploratory data analysis phase indicates that lexical richness is similar across genders, the findings reveal a significant disparity in the distribution of topic weights. Specifically, I find that female candidates exhibit a high degree of thematic ownership over a specialized rhetoric intersecting gender identity and labor struggle, whereas male discourse is defined by established institutional and administrative frameworks. These results could suggest that in a masculine-dominated political default, identity-signaling served as a critical strategic tool for minority female candidates to differentiate their manifestos.

Contents

1	Introduction	2
2	Literature Review	2
2.1	Gendered Political Rhetoric	2
2.2	Topic Modeling in Political Science	3
3	Data	3
3.1	Preprocessing and Filtering	3
3.2	Exploratory Data Analysis	3
3.3	Keyword Dominance and Ranking	4
4	Methodology	4
4.1	Vectorization and Feature Engineering	4
4.2	Non-Negative Matrix Factorization (NMF)	5
4.3	Model Selection and Stability	5
5	Results and Discussion	5

5.1	Thematic Ownership and the Identity Gap	7
5.2	Institutional Status as a Masculine Default	7
5.3	Limitations	7
5.4	Robustness Check: Sensitivity Analysis at $k = 10$	8
5.5	Comparison with Latent Dirichlet Allocation (LDA)	9
6	Conclusion	10

1 Introduction

Political communication is frequently analyzed through ideological shifts, yet the influence of gender on the structure of election rhetoric remains under-examined in late 20th-century France. Between 1973 and 1993, female candidates occupied a marginalized position within the French political space, representing only a small fraction of the total candidate pool. This demographic imbalance provided me with an opportunity to apply computational linguistics to determine whether the rigid structure of the election manifestos enforces a uniform rhetorical standard or allows for gendered differentiation. Following the logic of the "double burden" in political representation [1], I investigate the extent to which gender acts as a determinant for thematic focus.

I structured my methodology through three distinct analytical stages. Initially, I characterized the corpus statistically, establishing that despite the historical gender imbalance, the fundamental complexity of the language (measured via lexical richness) remains invariant between groups. I then utilized TF-IDF vectorization and Non-negative Matrix Factorization (NMF) to decompose the corpus into 15 latent topics. I chose NMF for its ability to isolate overlapping themes within a single document [2]. By mapping these topic weights back to candidate metadata, I quantified the "ownership" of specific political axes.

My findings challenge the notion of a gender-neutral political default. I demonstrate that female candidates disproportionately centered their rhetoric on a specific intersection of labor rights and gendered identity, utilizing these themes at a rate 5 times higher than their male counterparts. Conversely, I found that male rhetoric was heavily skewed toward institutional authority and regional administrative status, reflecting a "status quo" positioning afforded by their historical dominance of the political field. Finally, I address how the concentration of female candidates in Far-Left and Ecologist parties during this period informs these rhetorical clusters.

2 Literature Review

2.1 Gendered Political Rhetoric

Political science literature suggests that female candidates often face a "double bind": they must demonstrate competence in traditionally "masculine" domains like the economy and defense while simultaneously addressing "feminine" social issues to meet voter expectations [1]. Scholarly analysis of French legislative elections has noted that female representation during the 1970s and 1980s was largely restricted to the political periphery, primarily within Far-Left and emerging Ecologist parties [3]. This positioning often results in a "challenger" rhetoric that differs significantly from the institutionalized language used by incumbents, who were overwhelmingly male during this period [4]. My research builds on these theories by using unsupervised methods to quantify this "challenger" vs. "institutional" divide.

Beyond the French context, comparative studies have identified a persistent "gendered division of labor" in political discourse. For instance, research by Kahn and Gordon (1996) on U.S. senatorial campaigns found that female candidates are often stereotyped as experts in "compassion issues" such as education and healthcare, while being perceived as less competent in "hard" areas like economics and defense [5]. Similarly, Bauer (2020) demonstrates that female candidates often engage in "gendered signaling" to counteract these stereotypes or to mobilize specific voting blocs [6]. My research extends these findings by quantifying how these signals are mathematically represented in French manifestos.

2.2 Topic Modeling in Political Science

The application of Natural Language Processing (NLP) to political archives has transitioned from simple word counts to latent variable models. Latent Dirichlet Allocation (LDA) has been the traditional standard for identifying thematic shifts in government documents [7]. However, recent scholarship has highlighted the advantages of Non-negative Matrix Factorization (NMF) for political text, particularly for its ability to produce highly interpretable, non-probabilistic "parts-based" representations [2]. NMF is particularly suited for electoral manifestos, where a single document may cover disparate themes like fiscal policy and environmental protection. Unlike supervised classification, NMF allows me to discover these themes without a priori assumptions about what constitutes "male" or "female" topics, ensuring a more objective thematic extraction [8].

3 Data

The primary data source for this study is the *Archelec* archive, a digitized collection of French legislative election manifestos curated by Cevipof (Sciences Po). The original dataset consists of 21,167 documents spanning two decades of political history. To ensure the integrity of the gendered analysis, I implemented a cleaning pipeline to handle OCR noise and metadata inconsistencies.

3.1 Preprocessing and Filtering

The initial preprocessing involved the removal of standard stop-words, punctuation, and non-alphabetic characters using the *spaCy* library for French language processing. Given the historical nature of the documents, I applied lemmatization to reduce words to their base forms, ensuring that variations in gendered nouns and adjectives (e.g., *ouvrière* vs. *ouvrier*) were mapped to a common token where appropriate for thematic consistency.

I identified 836 documents where the candidate's gender was labeled as "non déterminé." To prevent noise in the comparative analysis, I dropped these rows entirely, resulting in a final clean dataset of 20,331 manifestos. This final sample maintains a significant gender imbalance, with 18,103 male candidates and 2,228 female candidates, reflecting the low levels of female representation in the French National Assembly during this period.

3.2 Exploratory Data Analysis

Before proceeding to topic modeling, I conducted a baseline statistical comparison to determine if linguistic complexity differed by gender. I utilized Type-Token Ratio (TTR) as a proxy for lexical richness. My analysis revealed an almost identical linguistic standard: male candidates exhibited a mean lexical richness of 0.841, while female candidates showed a mean of 0.842. This finding, as can be seen in Figure 1, is significant as it suggests that the "manifesto" genre imposes a formal linguistic constraint that transcends gendered author styles.

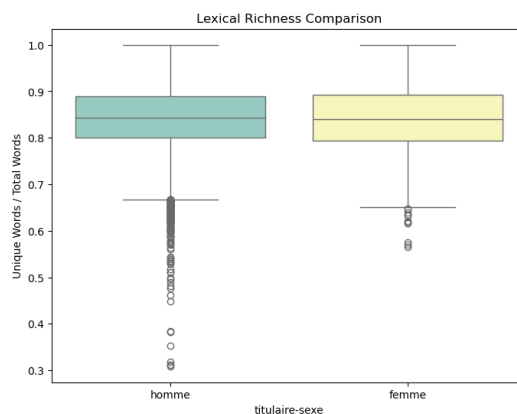


Figure 1: Distribution of Lexical Richness (TTR) by Gender. The overlapping densities confirm a high degree of linguistic similarity across the corpus.

3.3 Keyword Dominance and Ranking

A comparative rank analysis of the most frequent tokens provided the first evidence of thematic divergence. While both groups shared *travailleur* as their most frequent term, the secondary rankings revealed a sharp contrast. As illustrated in Figure 2, the token *femme* appeared as the second most frequent term in the female sub-corpus. In contrast, the male sub-corpus favored administrative and institutional terms such as *union*, *emploi*, and *maire*. This initial frequency-based evidence suggests that while male candidates relied on a "universal" or institutional rhetoric, female candidates were required to explicitly signal their gendered identity within their programs.

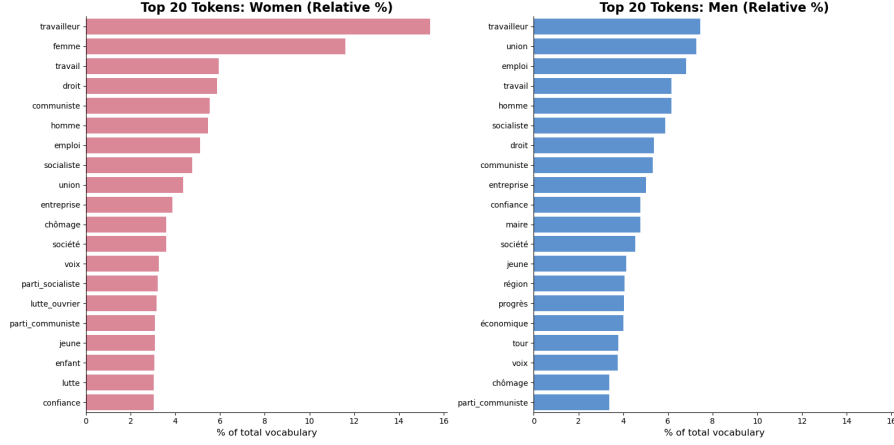


Figure 2: Comparative frequency of top tokens by gender.

To further visualize the high-level thematic weight of each sub-corpus, I generated word clouds based on relative token frequencies. As shown in Figure 3, these visualizations once again highlight the terminology and rhetoric differences between female and male candidates.

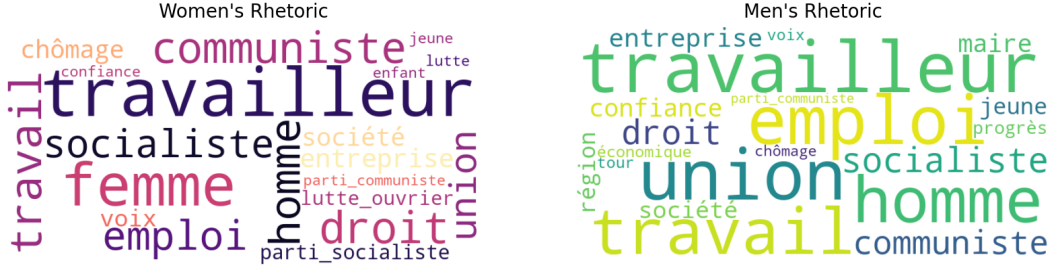


Figure 3: Comparative Word Clouds for female and male manifestos.

4 Methodology

I adopted an unsupervised machine learning approach to identify latent thematic structures within the corpus. This involved transforming the raw text into a numerical representation and applying matrix factorization to extract salient topics.

4.1 Vectorization and Feature Engineering

To prepare the corpus for modeling, I utilized Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. This transformation weights a term t in a document d as follows:

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \cdot \log \left(\frac{N}{\text{df}(t)} \right) \quad (1)$$

where $\text{tf}(t, d)$ represents the raw count of term t in document d , N is the total number of documents (20,331), and $\text{df}(t)$ is the number of documents containing term t . By applying this weighting, I prioritized distinctive political vocabulary while suppressing ubiquitous administrative terms. The resulting feature space was limited to terms with a minimum document frequency of 5 to ensure statistical stability.

4.2 Non-Negative Matrix Factorization (NMF)

I selected Non-Negative Matrix Factorization (NMF) as the primary decomposition model. Given a non-negative TF-IDF matrix $V \in \mathbb{R}^{n \times m}$, where n is the number of manifestos and m is the vocabulary size, NMF seeks to find two non-negative matrices W and H such that:

$$V \approx WH \quad (2)$$

In this framework, $W \in \mathbb{R}^{n \times k}$ represents the *document-topic matrix*, indicating the weight of each of the k topics in each document. The matrix $H \in \mathbb{R}^{k \times m}$ represents the *topic-word matrix*, containing the importance of each word within a given topic.

I chose the Frobenius norm as the objective function to minimize the reconstruction error:

$$\min_{W, H \geq 0} \|V - WH\|_F = \sqrt{\sum_{i,j} (V_{ij} - (WH)_{ij})^2} \quad (3)$$

Unlike Latent Dirichlet Allocation (LDA)¹, NMF's non-negativity constraint leads to a "parts-based" representation, which I found to be effective for isolating specific political sub-discourses.

4.3 Model Selection and Stability

The selection of the number of topics, k , is a critical hyperparameter. I determined the optimal k by evaluating the reconstruction error across a range of values [2, 40]. As illustrated in Figure 4, the error curve displays a characteristic "elbow" at $k = 15$. Beyond this point, adding further topics resulted in diminishing returns in error reduction and led to thematic fragmentation (e.g., the splitting of a single coherent ideology into redundant sub-topics). Consequently, I fixed $k = 15$ to achieve an optimal balance between the model's parsimony and semantic granularity.

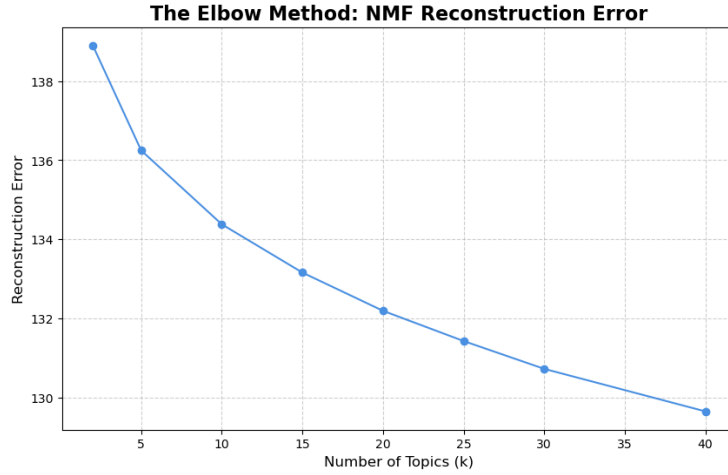


Figure 4: NMF Reconstruction Error vs. Number of Topics.

5 Results and Discussion

The NMF decomposition yielded 15 distinct thematic clusters, ranging from broad ideological platforms (e.g., Topic 14: Presidential Majority) to highly specific political identities. The semantic

¹ add here a footnote saying will later compare the results with my colleague's work

composition of these clusters, defined by the top ten tokens with the highest weights in matrix H , is visualized in Figure 5. This visualization provides the necessary qualitative context for interpreting the subsequent gendered distribution analysis.

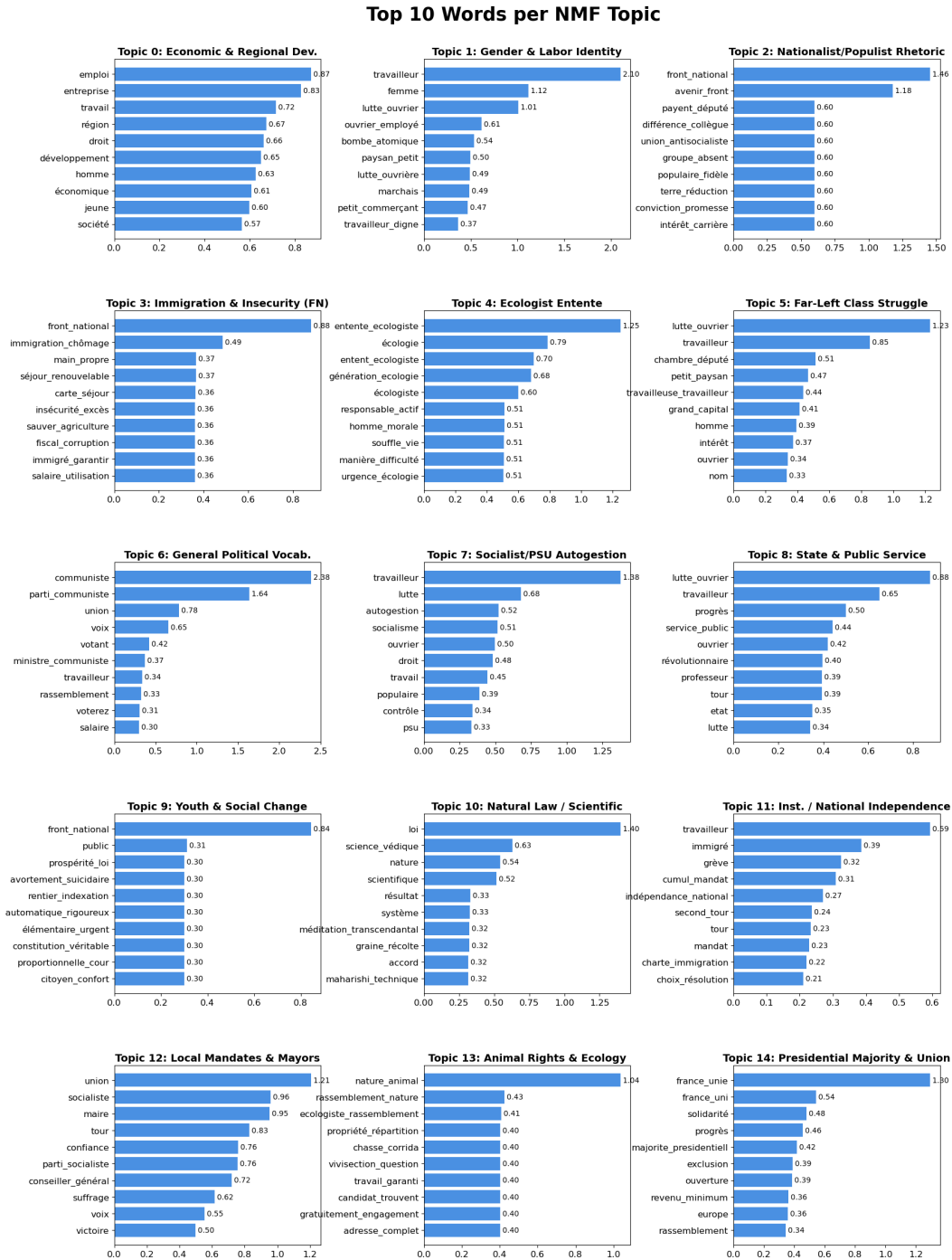


Figure 5: Thematic composition of the 15 NMF topics. Each subplot displays the top 10 tokens by weight.

To evaluate the gendered distribution of these themes, I mapped the document-topic weights from matrix W back to the candidate metadata and calculated the mean weight per topic for each gender.

5.1 Thematic Ownership and the Identity Gap

The primary finding of this research is the significant divergence in the utilization of Topic 1 (Gender and Labor Identity). As illustrated in the comparative heatmap in Figure 6, female candidates utilize this topic at a rate approximately 5 times higher than their male counterparts (Mean weight 0.0197 vs. 0.0043). As shown in the semantic breakdown in Figure 5, this topic is characterized by the co-occurrence of terms such as *travailleur*, *femme*, and *lutte_ouvrier*.

This disparity suggests that female candidates in this era did not rely on a "gender-neutral" political identity. While men utilized a "universal" rhetoric that largely ignored gender as a political category, women explicitly centered their identity within their programs to claim political space. This aligns with the idea that minority representatives must often perform additional rhetorical labor to justify their presence in a traditionally masculine apparatus.

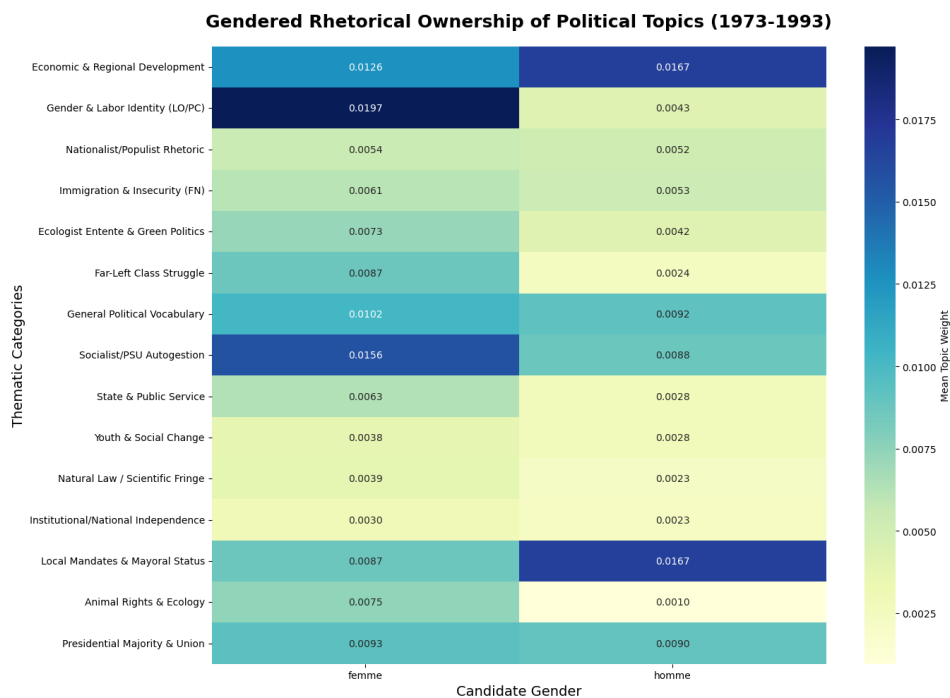


Figure 6: Mean Topic Weights by Gender.

5.2 Institutional Status as a Masculine Default

Conversely, male candidates exhibited higher mean weights in Topic 0 (Economic Development) and Topic 12 (Local Institutional Status). Topic 12 is particularly revealing, as it is anchored by tokens such as *maire* and *conseiller_général*. The male "ownership" of this topic (0.0167 vs. 0.0087 for women) reflects the historical reality of the era: men were more likely to hold multiple mandates and established local offices. Consequently, their rhetoric was rooted in the "status quo" and administrative experience, whereas women (often running as challengers or from the political periphery) had to adopt a more transformative and activist tone.

5.3 Limitations

A critical limitation of this analysis is the confounding influence of party affiliation. During the 1970s and 1980s, female candidacies were disproportionately concentrated within Far-Left parties (e.g., *Lutte Ouvrière*) and the burgeoning Ecologist movement. My results show that women score significantly higher in Topic 13 (Animal Rights and Ecology) and Topic 5 (Far-Left Struggle). It is therefore difficult to isolate whether the identified "gendered rhetoric" is a product of gender alone or a reflection of the specific ideological platforms of the parties that were most willing to field female candidates.

Furthermore, I acknowledge the structural ‘incumbency bias’ present in the archive. The male dominance of Topic 12 (Local Mandates) likely reflects historical access to political office rather than a linguistic preference; men spoke as incumbents because they held the seats, whereas women spoke as challengers because they were often excluded from the institutional default. Finally, while the NMF reconstruction error stabilized at $k = 15$, the qualitative labeling of these clusters remains an interpretive process. Future work could involve controlling for party labels and utilizing more recent, higher-quality digital archives to determine if these gendered rhetorical signals persist as women enter the political space.

5.4 Robustness Check: Sensitivity Analysis at $k = 10$

To validate the structural integrity of the identified themes, I performed a sensitivity analysis by collapsing the model to $k = 10$ topics (see Figures 7 and 8). The results confirm high thematic stability. Specifically, the ‘Gender & Class Identity’ cluster remained the primary differentiator, with female candidates utilizing this rhetoric at a rate approximately 4.6 times higher than male candidates.

This persistence suggests that the gendered gap is not necessarily due to model granularity, but it is a fundamental feature of the 1973–1993 political period. Furthermore, the higher male weight in Topic 0 (Institutional & Local Dev.) reinforces the finding that male rhetoric was consistently anchored in administrative status, whereas female rhetoric was centered more on identity-based signaling.

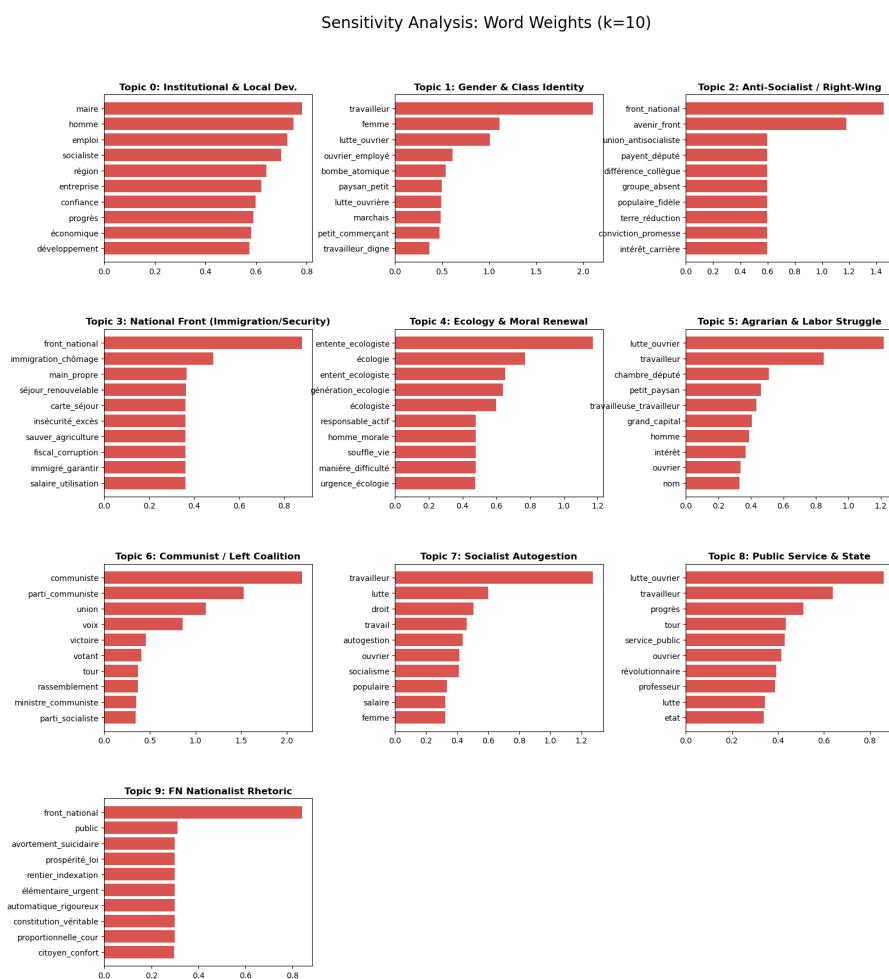


Figure 7: Sensitivity Analysis ($k = 10$): Word Weights per Topic.

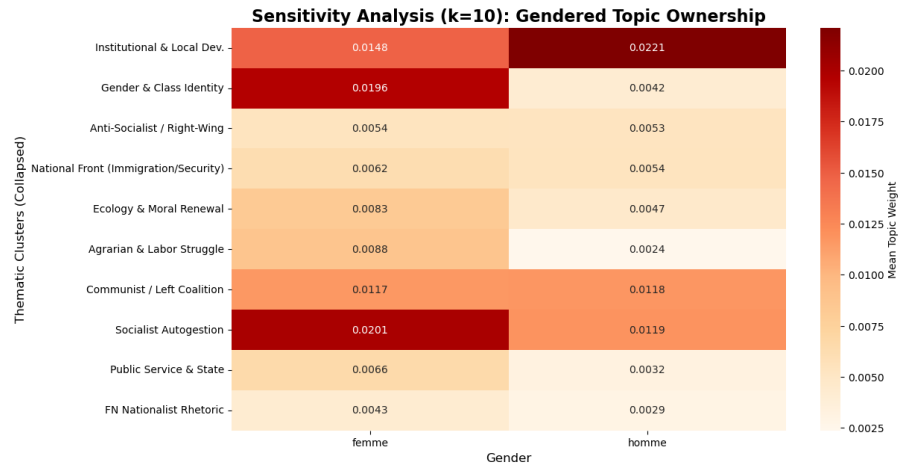


Figure 8: Sensitivity Analysis ($k = 10$): Mean Topic Weights by Gender.

5.5 Comparison with Latent Dirichlet Allocation (LDA)

To evaluate the efficacy of my NMF framework, I compared my results with an alternative decomposition conducted by my colleague² using Latent Dirichlet Allocation (see Figure 9). While both our models successfully identified a significant gendered divergence in identity-based rhetoric (with the "Gender & Labor Identity" cluster showing a roughly 5-fold increase in prevalence for female candidates in both frameworks), the LDA approach showed substantial "topic dominance" that I sought to avoid.

In the LDA model, a single "Economic & Regional Development" topic accounted for over 51% of the mean topic weight for male candidates and 28% for female candidates. In my view, this high concentration in a "catch-all" economic category tends to overshadow the more nuanced rhetorical shifts I observed in my NMF model. By contrast, I found that NMF provided a more balanced "parts-based" representation, allowing me to better separate institutional administrative language from specific ideological signaling. This comparison confirms to me that while the gendered signal is robust across different algorithms, NMF offered the superior thematic resolution necessary for my analysis of marginalized "challenger" discourse within this demographically imbalanced corpus.

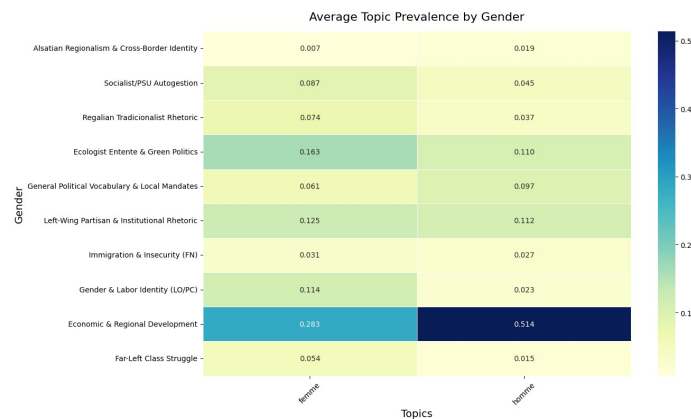


Figure 9: Comparative LDA Heatmap.

²The LDA analysis referenced in this section was conducted by Maria Micaela Linares Gomez as part of the course's final project.

6 Conclusion

In this project, I utilized NMF-based topic modeling to analyze a longitudinal archive of French electoral manifestos (1973–1993), moving beyond simple lexical counts to identify the latent thematic structures of gendered discourse. My analysis demonstrates that while linguistic complexity remains invariant between groups, political focus is deeply bifurcated. I found that female candidates disproportionately relied on "Identity and Labor" signaling, utilizing these themes at 5 times the rate of their male counterparts, in order to differentiate themselves within a marginalized political space.

By contrast, male rhetoric remained anchored in institutional stability and local administrative mandates, reflecting a "status quo" positioning due to historical dominance. The robustness of this gendered signal was confirmed through both sensitivity analysis and comparative modeling with LDA. Ultimately, this project provides a framework for understanding how minority voices deal with an institutional default, showing that the electoral manifesto are not neutral documents, but they reflect instead identity-based rhetorical struggle.

References

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